SUPPLEMENTARY MATERIAL

Rapid Adaptive Optical Recovery of Optimal Resolution over Large Volumes

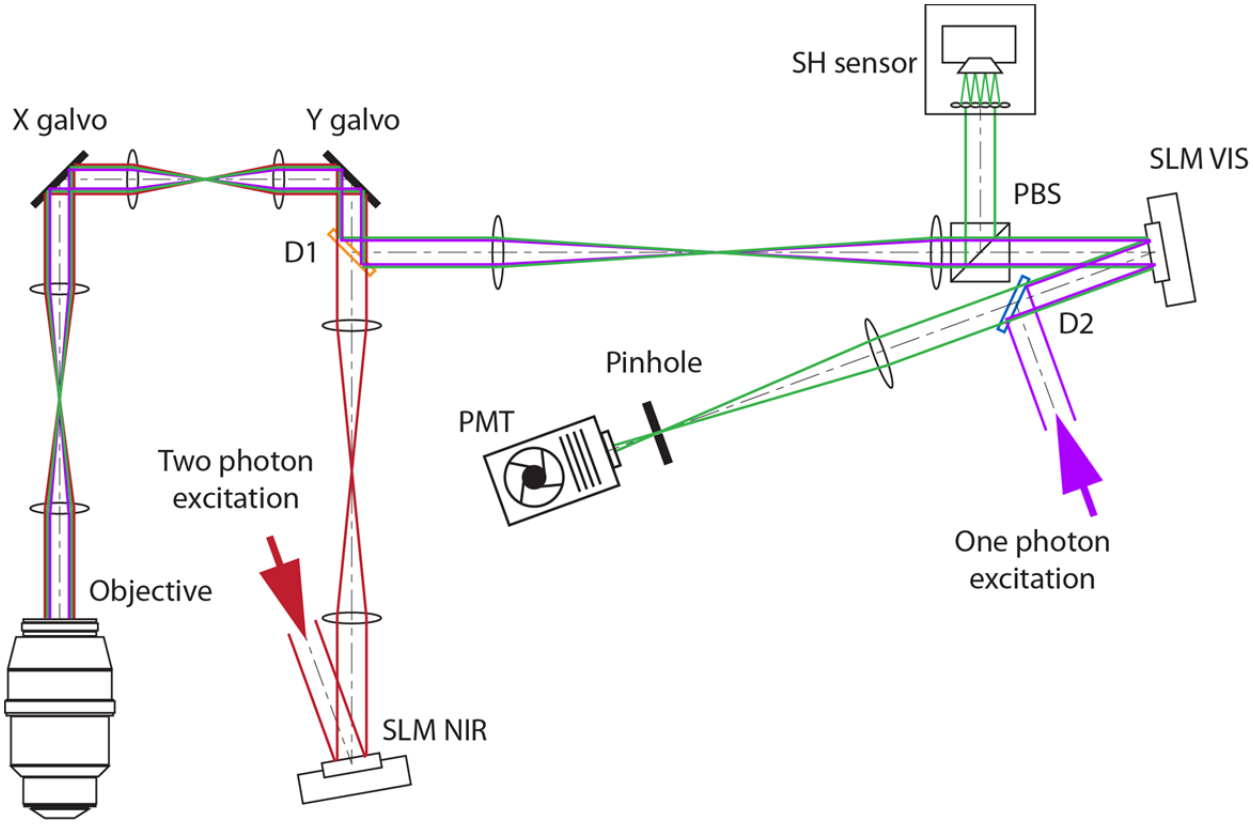
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Note: Supplementary Videos are available on the Nature Methods website.

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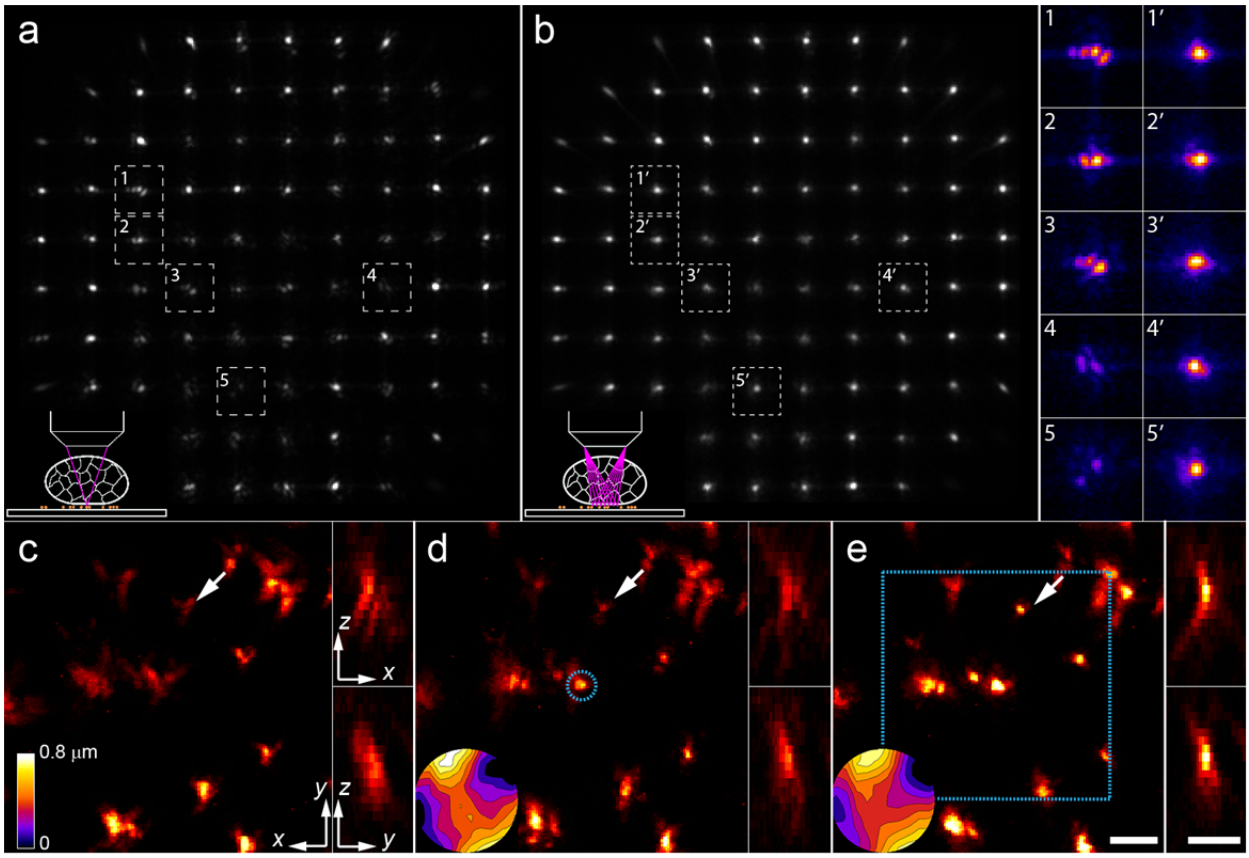
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Supplementary Figure 1| Simplified schematic of the combined two-photon / confocal / adaptive optical (AO) microscope.



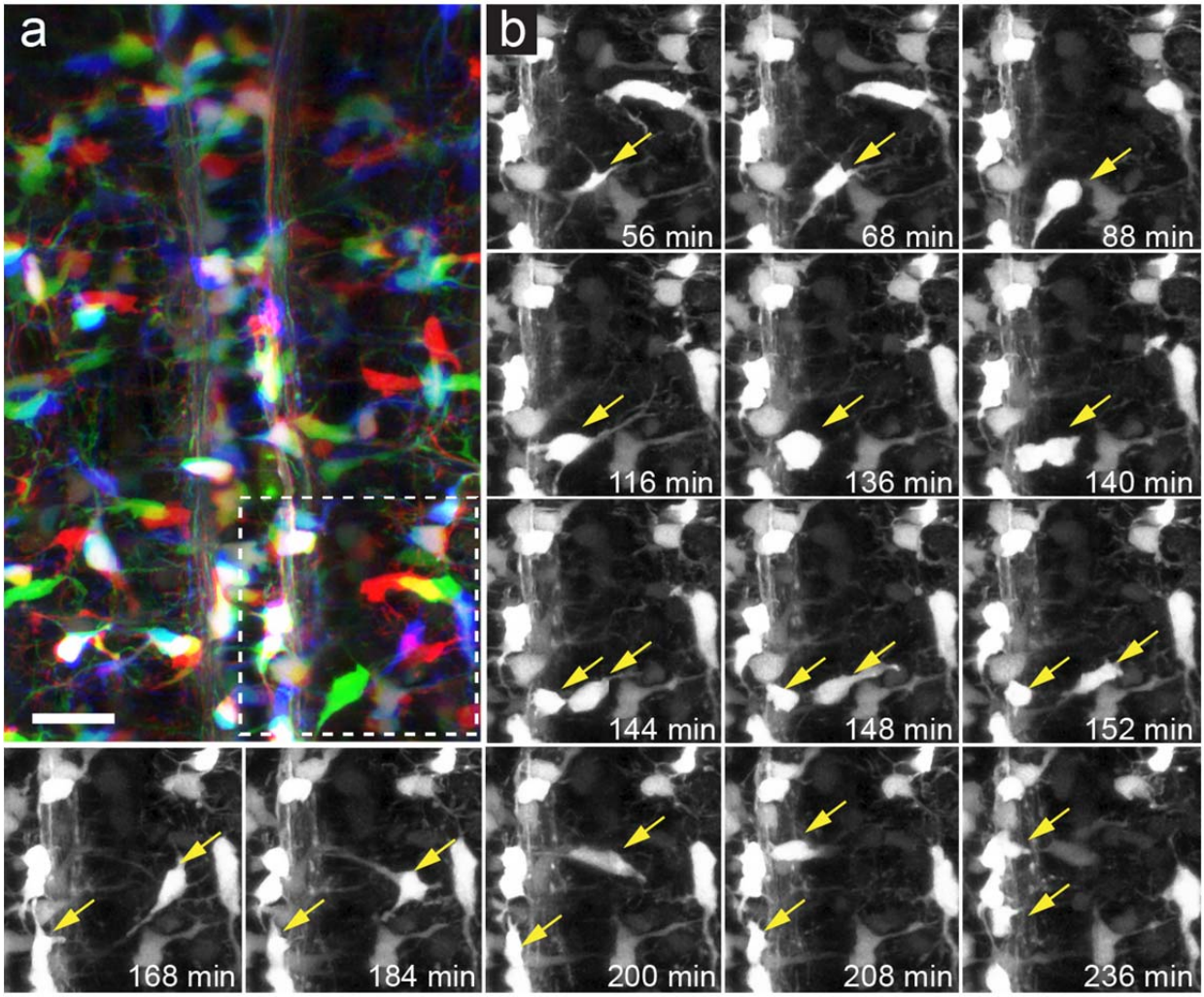
Near-infrared (NIR) pulsed excitation (red) is reflected from a NIR-responsive spatial light modulator (SLM NIR) and a pair of scanning galvanometer mirrors (X and Y galvos) before entering the rear pupil of a 1.1 NA long working distance water dipping objective. The SLM, both galvos, and the objective rear pupil are all mutually conjugate, so the phase pattern from the SLM is stationary at the rear pupil, even as the galvos scan the focused IR light laterally across the specimen. The fluorescence (green) thereby emitted and collected by the objective is reflected from the same galvos, separated from the NIR excitation with a dichroic beamsplitter (D1), and split into two orthogonal paths with a polarizing beamsplitter (PBS). One path enters a Shack-Hartmann (SH) sensor to determine the aberrated wavefront of the emission, and the other is reflected from an SLM responsive to visible light (SLM VIS) before being focused at a photomultiplier tube (PMT). SLM VIS and the front focal plane of the SH sensor are also each conjugate to the objective rear pupil. In the confocal mode, visible excitation light (purple) is reflected by a second dichroic beamsplitter (D2), SLM VIS, and Galvos X and Y before entering the objective. The emission then follows the same path as in the two photon mode, except that a pinhole is placed before the PMT at a location conjugate to the focus within the sample. In the two-photon mode, AO correction occurs simultaneously with image formation, and the measured wavefront is used to apply the same corrective pattern to both SLM NIR and SLM VIS. In the confocal mode, AO correction occurs sequentially: in each corrective subregion, the emission wavefront produced by scanning the two-photon focus over a fraction of the corrective subvolume (often a single plane) is first applied to both SLM NIR and SLM VIS, after which the IR excitation is turned off, the visible excitation turned on, and the entire subvolume imaged.

Supplementary Figure 2| Comparison of two-photon guide star based AO with and without de-scan.



(**a**) Spot pattern at the Shack-Hartmann (SH) wavefront sensor, for a guide star fixed at a fluorescent bead (blue circle in **d**), after passage through a *C elegans* embryo, ~3 hours post fertilization (inset in **a**). (**b**) Same, except after scanned excitation and de-scanned detection (inset in **b**) over a larger sub-region (blue square in **e**). Numbered spots in **a** and **b** are compared in columns at right (left column, fixed; right column, scan / de-scan). (**c**) Maximum intensity projection (MIP) wide field image of a field of 200 nm fluorescent beads through the embryo, without AO. MIP views of a specific bead (arrow) are also shown in the x-z (upper right) and y-z (lower right) planes. (**d**) Same, except with AO provided by a guide star fixed at the blue circle. Inset at lower left shows the wavefront calculated from the SH pattern in **a**. (**e**) Same, except with AO provided by a guide star scanned over the blue rectangle, and then de-scan of the emission before the SH sensor. Inset at lower left shows the wavefront calculated from the SH pattern in **b**. Note that de-scan provides a better correction not only for beads across the entire field of view, but also for the specific bead chosen as the target for the fixed guide star, since the local complexity of the aberration there leads to complex speckle patterns at the SH sensor and thus an inaccurate measurement of the wavefront. Scale bars, 2 μm. Scale is the same in **c**, **d** and **e**.

Supplementary Figure 3| Time lapse imaging of oligodendrocyte migration in the developing zebrafish hindbrain.



All frames are MIPs extracted from a 100 time point series (**Supplementary Video 2**) taken at 4 minute intervals over a 170 x 90 x 60 μm volume with AO and deconvolution in the two-photon mode, starting ~72 hours post-fertilization. Upper left: Multi-color overlay at t = 0 min (red), 80 min (green), and 160 min (blue) distinguishes motile oligodendrocytes from stationary ones. Time series: MIPs from the sub-region indicated by the white rectangle in the overlay, at the time points shown. Arrows show a cell entering the volume and then dividing, with both cells eventually migrating to the spinal cord region. Scale bar, 20 μm. Scale is the same in **a** and **b**.

Supplementary Figure 4| Zernike modes for the wavefront in Fig. 2c, near the zebrafish spinal cord.

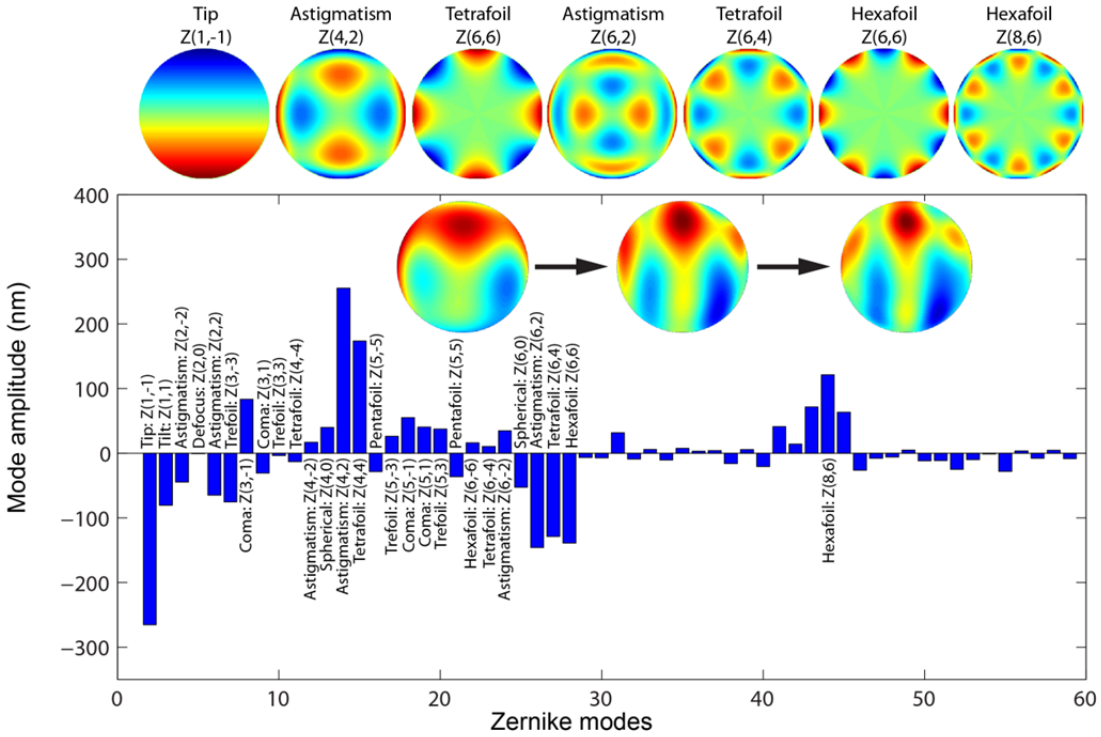
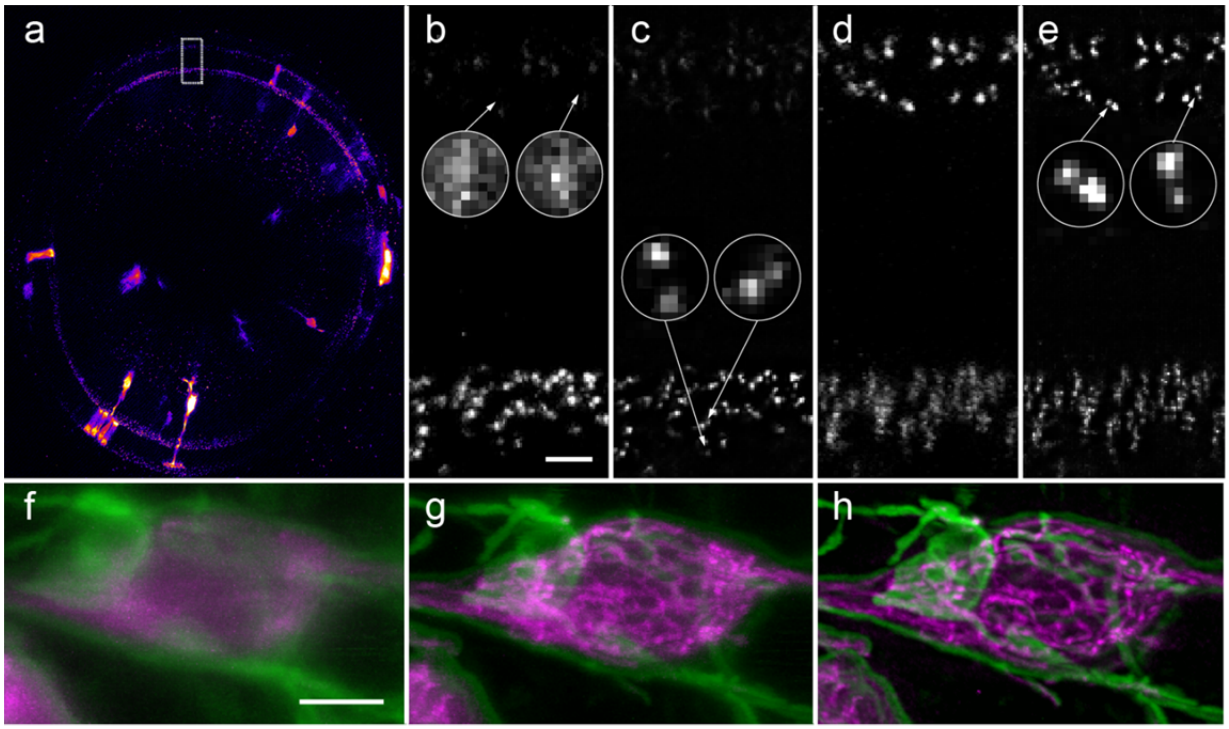


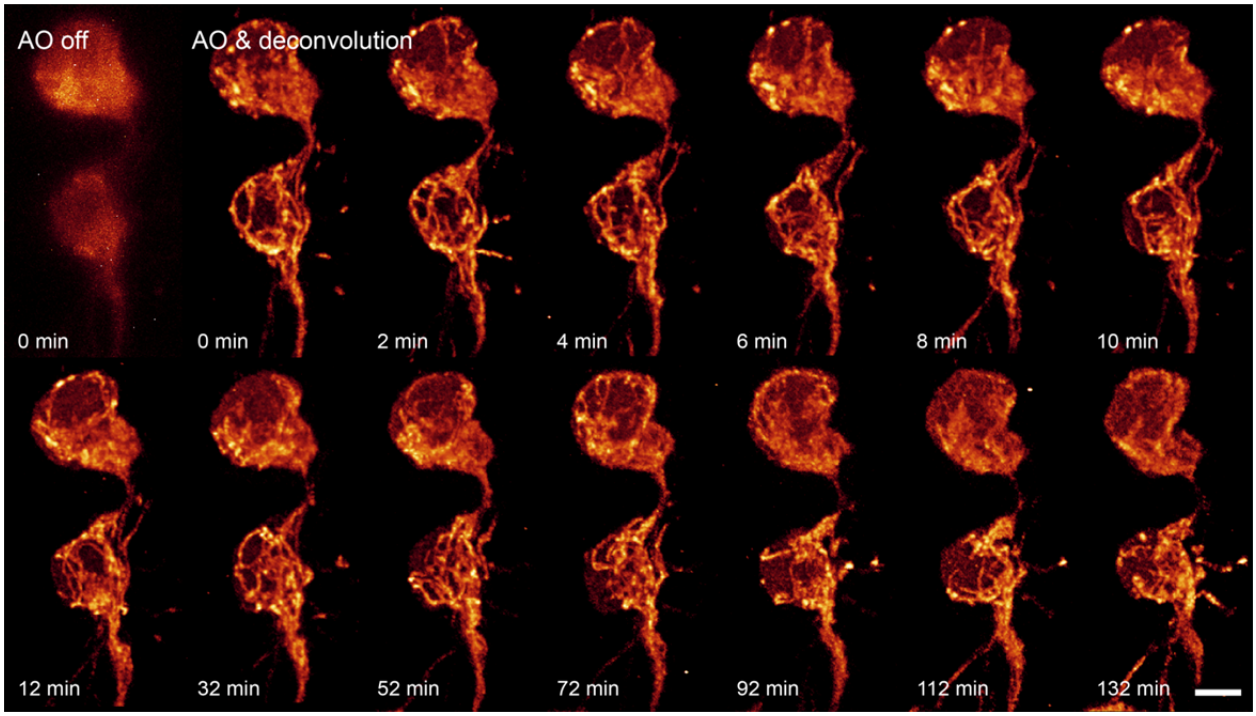
Table gives the relative amplitudes of the Zernike modes in a modal decomposition of the wavefront. Terms larger than $\lambda/10$ exist out to the 45th mode, demonstrating the need for a corrective device, such as a spatial light modulator, capable of generating such high order modes. Inset shows the wavefront as calculated with, from left to right, only the first 15, 28, or 45 modes. Panel at top shows the shapes of the seven strongest modes in the decomposition.

Supplementary Figure 5| Imaging sub-cellular structure with adaptive optics in the confocal mode.



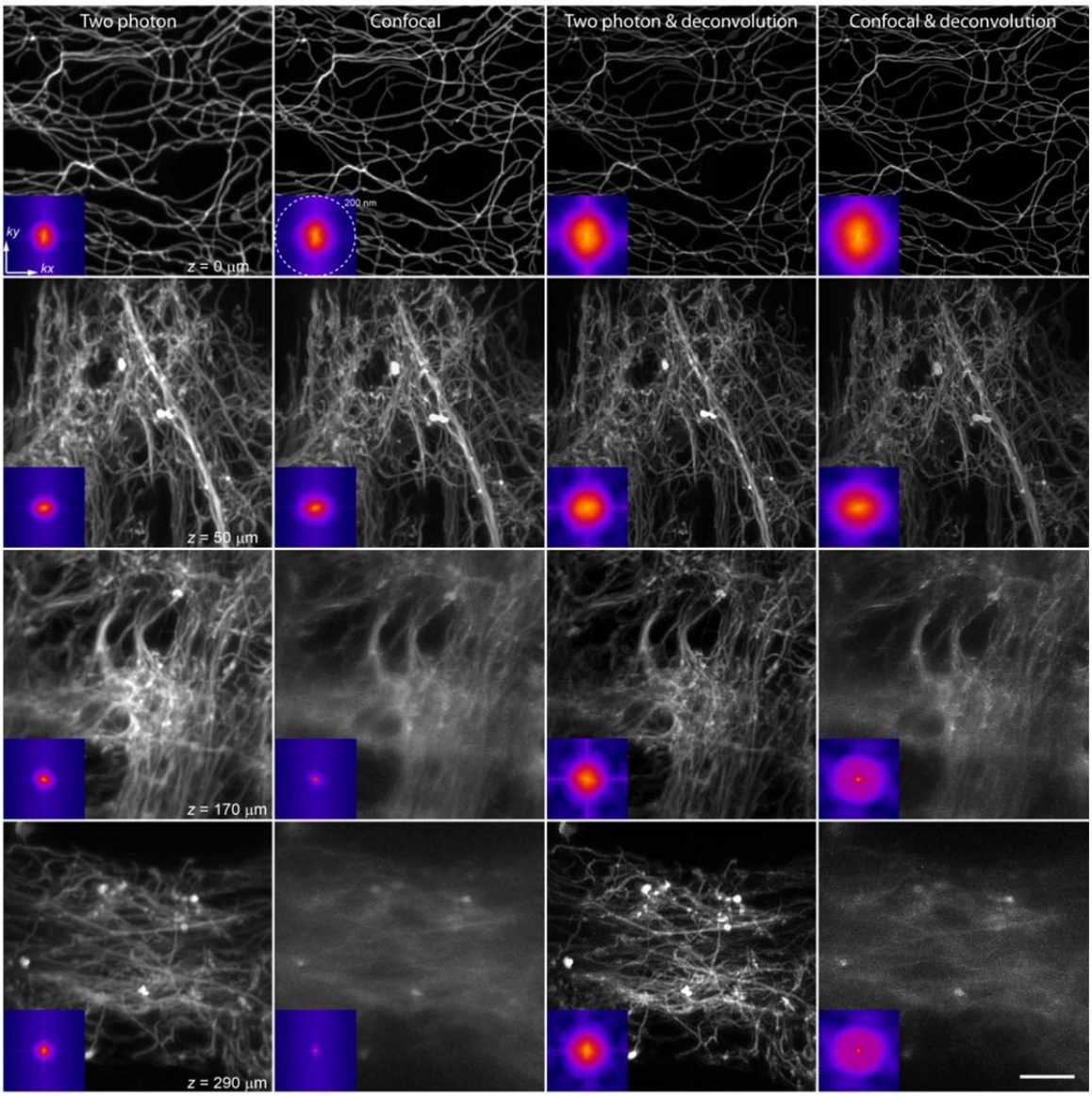
(**a**) MIP of a 15 μm-thick volume across the zebrafish eye, 5 days post fertilization, showing expression of centrin2-YFP concentrated in two bands. The outer band contains centrosomes of retinal photoreceptos, the inner band coincides with the outer plexiform layer and contains centrosomes of inner nuclear layer neurons. (**b**) Higher magnification of the region in the white box in a, without AO, showing reduced signal and resolution in the outer band, where the curvature of the eye leads to higher aberration. Scale bar, 2 μm. Scale is the same in **b**, **c**, **d** and **e**. (**c**) Same as **b**, except after deconvolution. The centrosomes in the inner band are resolved into centriole pairs, even without AO. (**d**) Same as **b**, except with AO applied to the centrosomes in the outer band. This correction degrades the resolution and signal at the inner band. (**e**) Same as **d**, except after deconvolution. Now the centrosomes in the outer band are resolved into centriole pairs. (**f**) Two color confocal MIP without AO of mitochondria (magenta) and the plasma membrane (green) of a cell ~150 μm deep in the zebrafish hindbrain. Scale bar, 5 μm. Scale is the same in **f**, **g** and **h**. (**g**) Same, except with AO. (**h**) Same, except after AO and deconvolution. See also **Supplementary Video 5**.

Supplementary Figure 6| Time lapse imaging of mitochondrial trafficking in neurons ~150 μm deep in the zebrafish brain.



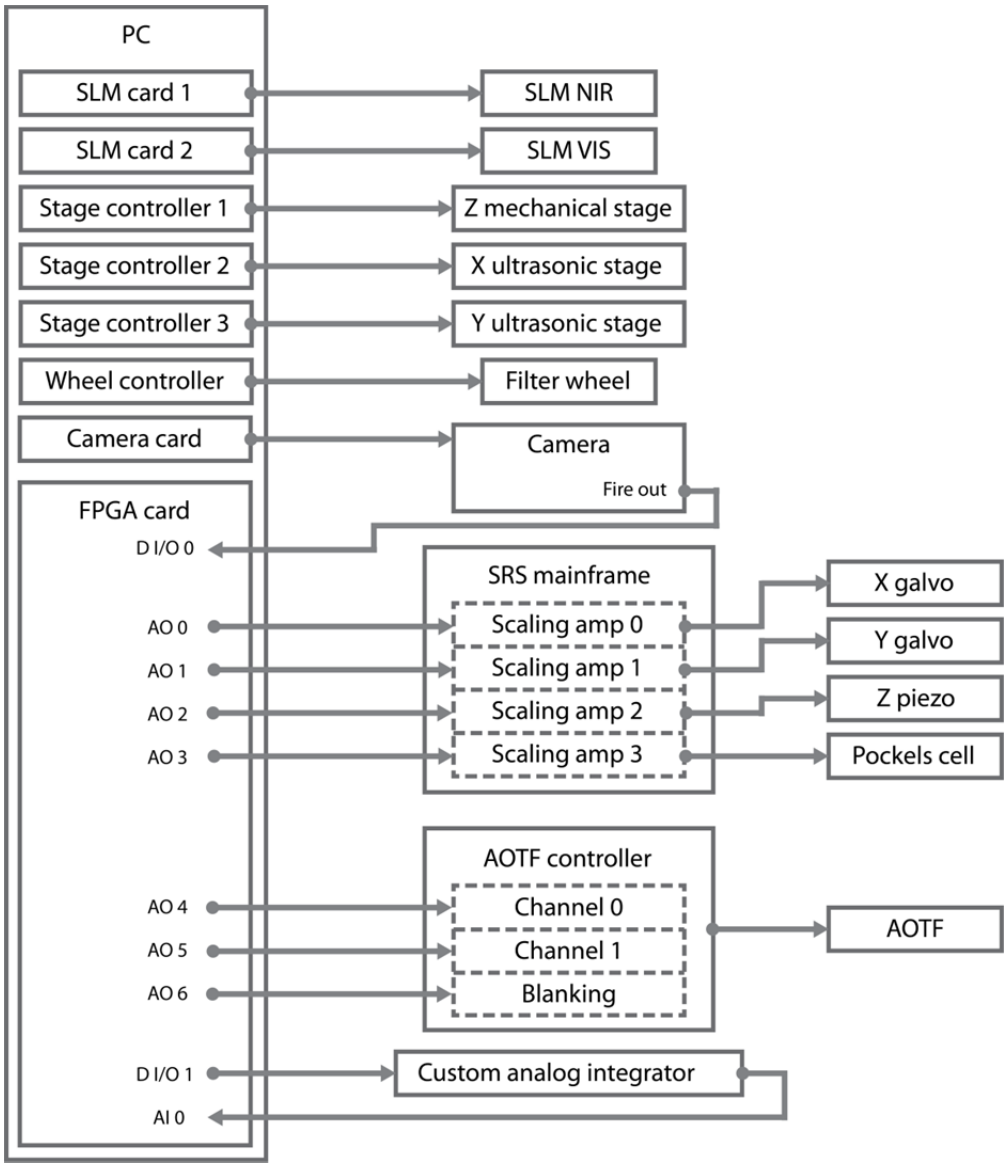
All frames are MIPs extracted from a 82 time point series (**Supplementary Video 6**) taken at 2 minute intervals over a 20 x 40 x 18 μm volume with AO and deconvolution in the confocal mode, starting ~4 days post-fertilization. Scale bar, 10 μm.

Supplementary Figure 7| Comparison of adaptive optics in the two-photon and confocal imaging modes.

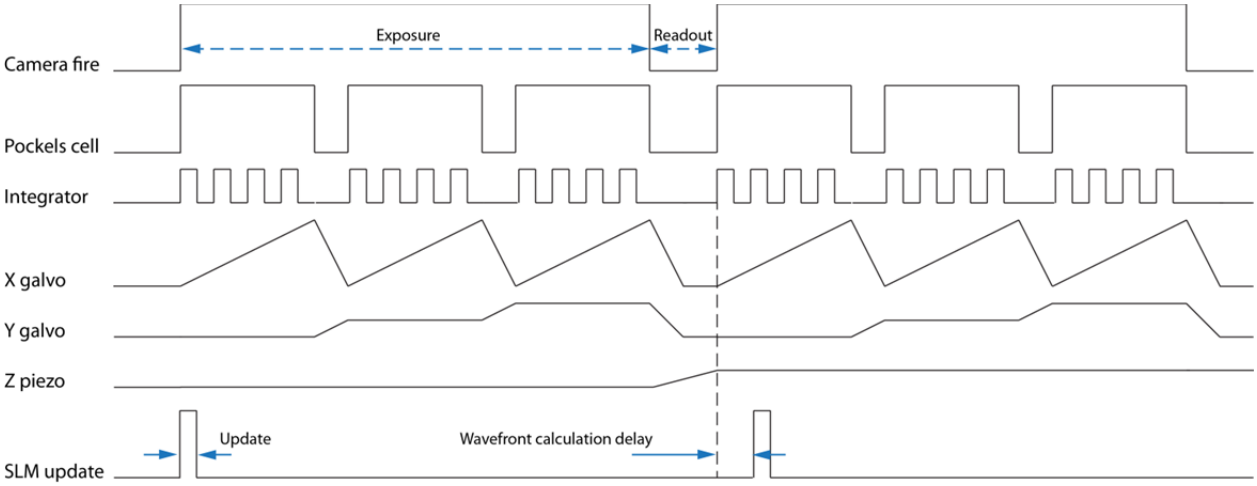


All images are x-y MIPs of 50 x 50 x 10 μm volumes showing membrane-labeled neurons in the brain of a zebrafish, ~72 hours post fertilization. Columns, from left to right: two-photon mode with AO only; confocal mode with AO only; two-photon mode with AO and deconvolution; and confocal mode with AO and deconvolution. Each row shows MIPs in the same region and at the depths shown at left. Each inset shows an x-y summation of the spatial frequency content of the corresponding image. Both real and frequency space representations in the top two rows show the ability of both imaging modes, with AO and deconvolution, to recover information out to the diffraction limit to ~50 μm depth. In this regime, the confocal mode, using visible light, outperforms the infrared-based two-photon mode. However, at greater depths (rows 3, 4), the strong scattering of visible light makes it impossible to recover all but the lowest spatial frequencies in the confocal mode, whereas the weaker scattering of infrared light permits the recovery of most spatial frequencies, even down to 290 μm. Scale bar, 10 μm. Scale is the same in all panels.

Supplementary Figure 8| Instrument control architecture.



Supplementary Figure 9| Timing diagrams and control waveforms for AO in the two photon imaging mode.



Supplementary Table 1| Acquisition parameters for images in the main text.

Image property	Figure 1a	Figure 1b,c and d	Figure 2a
Modality	two photon	two photon	two photon
Sample	Zebrafish (~3 dpf) roy^{a9}; gmc604Et; gmc930Tg	Zebrafish (~3 dpf) roy^{a9}; gmc604Et; gmc930Tg	Zebrafish (~3 dpf) Tg(β-actin:mgfp)
Fluorescent targets	YFP, sparse set of neural membranes	YFP, sparse set of neural membranes	EGFP, ubiquitous membrane expression
Excitation wavelength (nm)	960	960	920
Voxel volume (nm^3)	120x120x350	120x120x350	120x120x350
Image volume (μm^3)	240x240x270	50x50x17.85	600x750x6
Pixel rate (pixels/s)	100k	100k	200k
Number of subunits	8x8x18	2x2x1	20x26x1
Subunit size (μm^3)	30x30x17.85	30x30x17.85	30x30x6
Number of AO corrections per subunit	17	17	6
Gamma	0.6	1.0	0.8
Corresponding supplementary video	Supplementary Video 1	Supplementary Video 1	Supplementary Video 3

Supplementary Table 1 (cont'd)| Acquisition parameters for images in the main text.

Image property	Figure 2b,c and d	Figure 3
Modality	two photon	confocal
Sample	Zebrafish (~3 dpf) Tg(–4.9sox10:eGFP)	Zebrafish (~3 dpf) Tg(–4.9sox10:eGFP) Tg(8.4neurog1:nRFP)
Fluorescent targets	cytosolic EGFP, subset of oligodendrocytes	cytosolic EGFP, subset of oligodendrocytes; nuclear RFP in neurons
Excitation wavelength (nm)	920	488/561
Voxel volume (nm^3)	120x120x350	100x100x300
Image volume (μm^3)	50x50x10	40x40x200
Pixel rate (pixels/s)	100k	60k
Number of subunits	1x1x1	1x1x25
Subunit size (μm^3)	50x50x10	40x40x9
Number of corrections per subunit	1	1
Gamma	0.6	0.5
Corresponding supplementary video	none	Supplementary Video 4

Supplementary Table 2| Acquisition parameters for images in supplementary figures.

Image property	Supplementary Figure 2c, d and e	Supplementary Figure 3	Supplementary Figure 5a
Modality	wide field	two photon	confocal
Sample	200 nm fluorescent beads underneath a C.elegans embryo (~3 hpf)	Zebrafish (~3 dpf) Tg(–4.9sox10:eGFP)	Zebrafish (~5 dpf) s1101t-Gal4 x UAS-memCerulean, UAS-centrin2YFP
Fluorescent targets	yellow-green beads (Invitrogen)	cytosolic EGFP, subset of oligodendrocytes	centrosomes targeted YFP in retinal neurons
Excitation wavelength (nm)	488	920	514
Voxel volume (nm^3)	100x100x300	200x200x800	300x300x1000
Image volume (μm^3)	15x15x6	120x180x50	300x300x10
Pixel rate (pixels/s)	50k	140k	100k
Number of subunits	1	4x3x1	1
Subunit size (μm^3)	15x15x6	31x61x50	300x300x10
Number of corrections per subunit	1	30	N.A.
Gamma	1.0	0.6	1.0
Corresponding supplementary video	none	Supplementary Video 2	none

Supplementary Table 2 (cont'd)| Acquisition parameters for images in supplementary figures.

Image property	Supplementary Figure 5b, c, d and e	Supplementary Figure 5f, g and h	Supplementary Figure 6
Modality	confocal	confocal	confocal
Sample	Zebrafish (~5 dpf) s1101t-Gal4 x UAS-memCerulean, UAS-centrin2YFP	Zebrafish (~3 dpf) HuC-Gal4 x UAS-mitoCFP/memYFP	Zebrafish (~4 dpf) HuC-Gal4 x UAS-mitoCFP/memYFP
Fluorescent label	centrosomes targeted YFP in retinal neurons	mitochondrial CFP, membrane YFP in a subset of neurons	mitochondrial CFP in a subset of neurons
Excitation wavelength (nm)	514	440/514	514
Voxel volume (nm^3)	100x100x300	100x100x300	100x100x300
Image volume (μm^3)	12x28x15	25x15x15	20x40x18
Pixel rate (pixels/s)	80k	50k	40k
Number of subunits	1x1x1	1x1x1	1x1x1
Subunit size (μm^3)	12x28x15	25x15x15	20x40x18
Number of corrections per subunit	1	1	1
Gamma	1.0	green: 0.3 magenta: 0.6	1.0
Corresponding supplementary video	none	Supplementary Video 5	Supplementary Video 6

Supplementary Table 2 (cont'd)| Acquisition parameters for images in supplementary figures.

Image property	Supplementary Figure 7
Modality	confocal and two photon
Sample	Zebrafish (~3 dpf) roy^{a9}; gmc604Et; gmc930Tg
Fluorescent target	YFP, sparse set of neural membranes
Excitation wavelength (nm)	514/960
Voxel volume (nm^3)	100x100x300
Image volume (μm^3)	50x50x10
Pixel rate (pixels/s)	83k
Number of subunits	1
Subunit size (μm^3)	50x50x10
Number of corrections per subunits	1
Gamma	1.0
Corresponding supplementary video	none